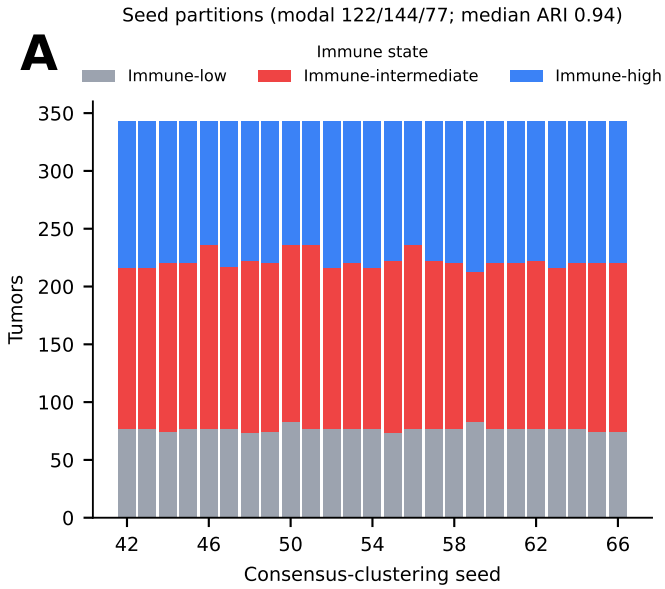
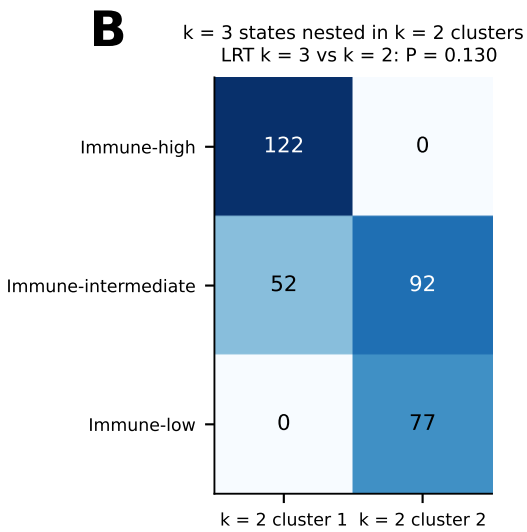
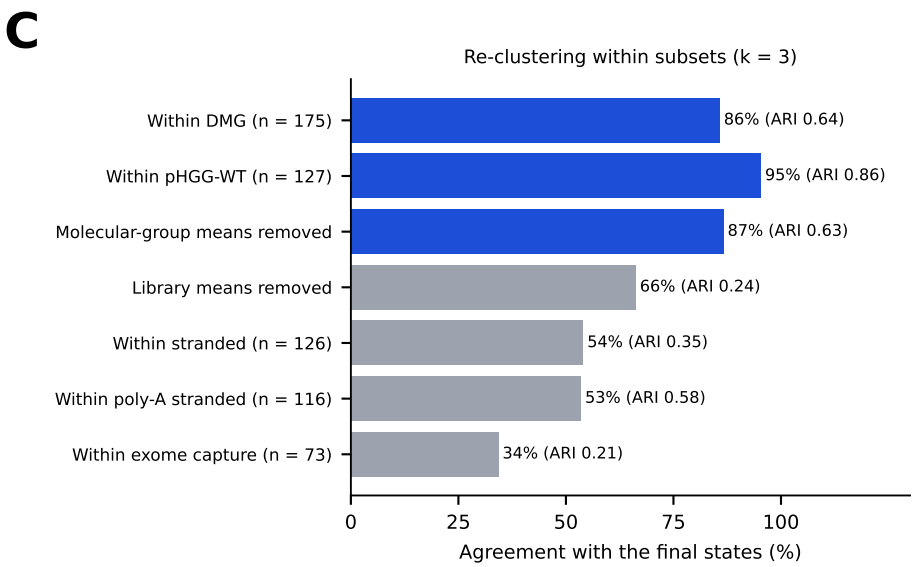
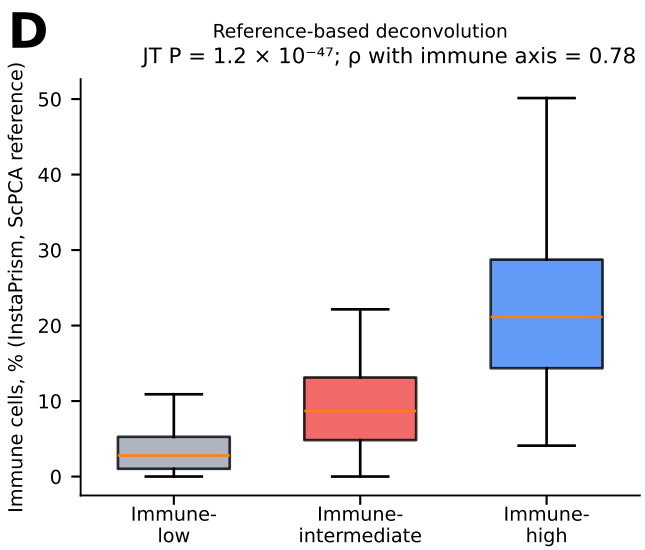
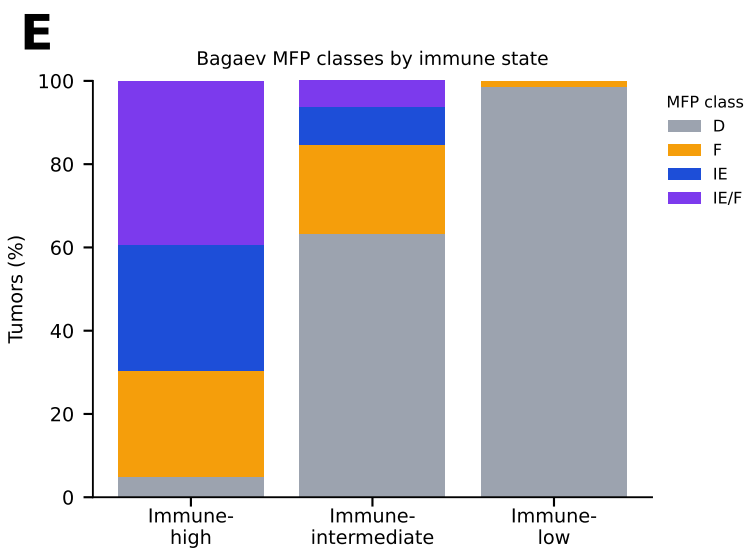
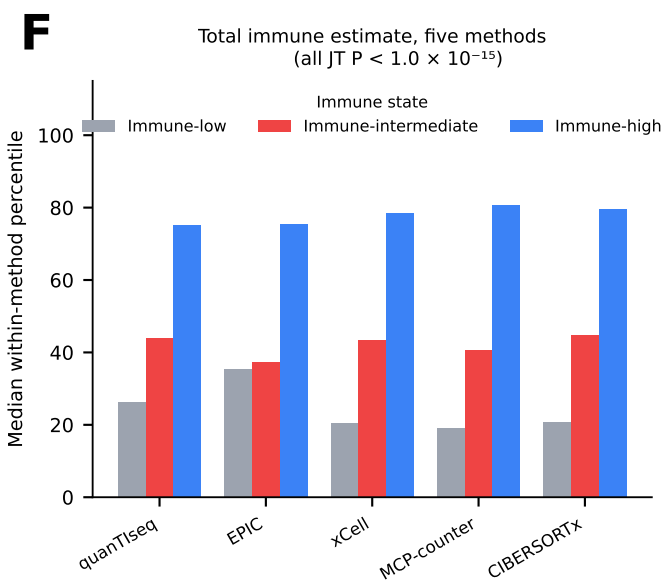

A
Seed partitions (modal 122/144/77; median ARI 0.94)
Immune state
Immune-low
Immune-intermediate
Immune-high
Tumors
Consensus-clustering seed


B
k = 3 states nested in k = 2 clusters
LRT k = 3 vs k = 2: P = 0.130
Immune-high
Immune-intermediate
Immune-low
122
0
52
92
0
77
k = 2 cluster 1
k = 2 cluster 2


C
Re-clustering within subsets (k = 3)
Within DMG (n = 175)
86% (ARI 0.64)
Within pHGG-WT (n = 127)
95% (ARI 0.86)
Molecular-group means removed
87% (ARI 0.63)
Library means removed
66% (ARI 0.24)
Within stranded (n = 126)
54% (ARI 0.35)
Within poly-A stranded (n = 116)
53% (ARI 0.58)
Within exome capture (n = 73)
34% (ARI 0.21)
Agreement with the final states (%)


D
Reference-based deconvolution
JT P = 1.2 × 10⁻⁴⁷; ρ with immune axis = 0.78
Immune cells, % (InstaPrism, ScPCA reference)
Immune-low
Immune-intermediate
Immune-high


E
Bagaev MFP classes by immune state
MFP class
D
F
IE
IE/F
Tumors (%)
Immune-high
Immune-intermediate
Immune-low


F
Total immune estimate, five methods
(all JT P < 1.0 × 10⁻¹⁵)
Immune state
Immune-low
Immune-intermediate
Immune-high
Median within-method percentile
quanTIseq
EPIC
xCell
MCP-counter
CIBERSORTx